LAURA BASSI SCHOLARSHIP
2026/27 APPLICATION FORM

EDITING PRESS

641 N BROAD ST, MIDDLETOWN, DE 19709
WWW.EDITING.PRESS

FIRST NAME	LAST NAME	E-MAIL
EUN SUNG	JANG	jangstar@ajou.ac.kr

INSTITUTION	DEPARTMENT	DEGREE OR POSITION
Ajou University	Artificial Intelligence Convergence	PhD Candidate

In 250 words or less, describe your manuscript's intended purpose. You may use this space to offer a general description of your project or an outline of your argument as it may appear in your abstract or introduction.

My current journal manuscript, titled "Design of Dual Polarized Distributed FMCW Radar System with High Gain Lens Array Antenna for Through the Wall Human Detection Application," presents the development of a compact radar platform designed to detect human presence and respiration behind concrete walls. The purpose of this manuscript is to demonstrate how combining dual polarization radar sensing, distributed radar architecture, and a high gain lens array antenna can significantly improve the detection of weak human signals in challenging indoor environments.

Through the wall human detection is difficult because radar signals are severely attenuated by building materials and often masked by strong clutter reflections and multipath propagation. Conventional FMCW radar systems typically rely on single polarization measurements or large multi channel configurations, which either limit detection performance or increase system complexity and deployment cost.

To address these limitations, the manuscript proposes a distributed FMCW radar system consisting of two spatially separated radar nodes operating with orthogonal polarizations. The system incorporates a compact high gain dual polarized lens array antenna and a polarization synthesis algorithm that coherently combines received signals to improve the signal to noise ratio. The manuscript presents the radar system architecture, antenna design, signal processing framework, and both simulation and experimental validation.

Overall, the purpose of this work is to demonstrate that a compact dual polarized distributed radar system can achieve reliable through the wall human detection while maintaining practical system complexity and deployment flexibility.

In 250 words or less, describe how your field of study stands to benefit from your research.

Radar based human sensing has become an important research area in applications such as smart environments, healthcare monitoring, security systems, and disaster response. However, detecting human presence in cluttered environments remains a challenging problem because the radar reflections from the human body are often weak and easily masked by environmental clutter.

This research contributes to the radar sensing field by introducing a dual polarized distributed FMCW radar framework that improves the detectability of weak human signals. While many existing radar sensing studies focus on centralized radar configurations or single polarization measurements, the combined use of distributed radar nodes and polarization diversity has received comparatively limited attention. By exploring this approach, the study provides new insights into how polarization information can enhance target detection and signal interpretation in radar systems.

The results of this research may contribute to improved radar system design strategies, particularly for applications requiring reliable detection of small or weak targets. The findings may also support future developments in distributed sensing architectures, where multiple radar sensors cooperate to observe complex environments.

Beyond theoretical contributions, the proposed approach may also have practical implications for emerging sensing technologies. Improved human detection capabilities could benefit applications such as search and rescue operations, through wall monitoring, and non contact human activity sensing.

Overall, this research expands the understanding of how polarization diversity and distributed radar sensing can be integrated to improve radar based human detection performance.

SELECT THE ROUND OF FUNDING FOR WHICH YOU ARE APPLYING:

☑	SPRING 2026	DEADLINE:	8 MARCH 2026
		RESULTS:	10 APRIL 2026
☐	SUMMER 2026	DEADLINE:	12 JULY 2026
		RESULTS:	24 JULY 2026
☐	WINTER 2026	DEADLINE:	15 NOVEMBER 2026
		RESULTS:	4 DECEMBER 2026

EUN SUNG JANG — SIGNATURE OF APPLICANT

2026.03.06 — DATE

Applicants are required to submit this application form along with their curriculum vitae through the application portal by the relevant deadline. For more information about the Laura Bassi Scholarship and access to the application portal, please visit https://editing.press/bassi.

To help defray the Scholarship's administrative costs, applicants are subject to a voluntary USD 10.00 fee. **All applicants who are unable to pay the application fee are welcome to take advantage of the fee waiver option on the application portal.** If you wish to pay the application fee in a non-USD currency, please consult the Scholarship FAQ.